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A nurse-led self-efficacy–based hybrid diabetic foot self-management education program for individuals with diabetes: a randomized controlled trial

Emine Çelik Polat¹, Ayşe Arıkan Dönmez^{1*} and Erman Çakal²

Abstract

Aim This study aimed to examine the effect of a nurse-led self-efficacy–based hybrid diabetic foot self-management education program on diabetic foot knowledge, foot care behaviors, adaptation to chronic illness, and self-efficacy among adults with type 2 diabetes mellitus (T2DM).

Design This study was a two-arm, single-center and single-blind, randomized controlled trial.

Methods The study sample consisted of 48 adults with T2DM assigned to the intervention ($n = 25$) and control ($n = 23$) groups using block randomization. The study was conducted at a tertiary hospital outpatient clinic in Ankara. The intervention group received a nurse-led self-efficacy–based hybrid foot self-management education program for eight weeks. This was a three-stage program comprising face-to-face sessions, WhatsApp-mediated interactions, and educational content provided through a YouTube channel, and it included a follow-up component. Baseline and follow-up data were collected at baseline, week 4, and week 8 using the Diabetic Foot Knowledge Scale (DFKS), Foot Care Behavior Scale (FCBS), Adaptation to Chronic Illness Scale (ACIS), and Diabetic Foot Care Self-Efficacy Scale (DFCSES). Data were analyzed using descriptive statistics and Repeated Measures ANOVA.

Results Significant Group $\times$ Time interaction effects were found for DFKS ($F = 37.50, p < 0.05$), FCBS ($F = 12.8, p < 0.05$), and DFCSES ($F = 3.87, p < 0.05$), indicating greater improvement in the intervention group compared to the control group. No significant Group $\times$ Time differences were found in ACIS ($F = 0.071, p > 0.05$).

Conclusion The self-efficacy-based, nurse-led hybrid diabetic foot self-management education program significantly improved diabetic foot knowledge, foot care behaviors, and self-efficacy in adults with T2DM. These findings support incorporating this education program into routine diabetes management to promote sustainable self-care and reduce the risk of diabetic foot complications.

Patient or public contribution Patients and public were not involved in this study. This study was registered on ClinicalTrials.gov (Ref. No: NCT05974592, Registered on July 26, 2023).

Keywords Diabetes mellitus, Diabetic foot, Self-management education, Nurse-led intervention

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Introduction

Diabetic foot is one of the most serious complications of diabetes, arising from the destructive interaction of peripheral neuropathy, peripheral artery disease (PAD), and recurrent trauma, and complicated by ischemia and infection [1]. Prolonged hyperglycemia causes sensory, motor, and autonomic neuropathy, leading to the loss of protective sensory perception, impaired foot biomechanics, and increased skin fragility [2]. PAD compromises tissue perfusion, resulting in diminished oxygenation and extended healing times [1]. Hyperglycemia-induced immune system impairment and chronic inflammation exacerbate the deterioration of tissue integrity. The interplay of these systems establishes a “vicious cycle” of ulceration, infection, and extended healing, markedly elevating the probability of outcomes that could result in lower limb amputation [1, 2].

According to the International Working Group on the Diabetic Foot (IWGDF) guideline, diabetic foot prevention relies on early identification of foot at risk, regular control and assessment of the foot, patient, family, and healthcare professional education, appropriate footwear, and treatment of risk factors for diabetic foot ulcers (DFUs) [2]. It is well-known that education for people with diabetes mellitus (DM) in clinical settings is often constrained to topics such as DM, insulin and oral anti-diabetic use, nutrition, physical activity, etc., even though DFUs are a complication that can be prevented with thorough and organized education [3]. Patient education is believed to play a role in preventing DFUs by assisting in the assessment of risk factors, and self-care knowledge, skills, and behaviors [4].

It is noteworthy that the research results in literature focus on the fact that diabetic foot care behaviors of individuals with DM are inadequate or that adequate foot care is not performed [5, 6]. A critical study in Türkiye found that 90.9% of patients with DM did not receive any education on foot care and 54.5% did not perform routine foot care [7]. After conducting a thorough examination of these data, it is evident that individuals require specialized education from healthcare professionals regarding DM and related complications.

Diabetic foot is a significant issue that necessitates the patient to perform complex care activities and provide self-management throughout their lifetime [8]. Structured diabetes self-management education programs (DSMEP) enable individuals with diabetes or those at risk to acquire the necessary knowledge and skills to manage their disease and develop behavioral changes related to self-management [9]. It has been shown that DSMEP forms the basis of care for all individuals with type 2 diabetes mellitus (T2DM) in whom improved health outcomes are targeted [9]. The literature emphasizes that nurses play a key role as healthcare professionals in

educational interventions for diabetic foot, in screening high-risk individuals, and in providing healthcare, particularly in preventing foot ulcers and lower extremity amputations [10, 11]. A study examining the effects of self and family management support programs to improve clinical outcomes in DFUs found significant improvements in self-management, family support, HbA1C levels, and wound size [12]. Although several randomized trials suggest potential benefits of self-care-focused interventions for glycemic and diabetes-related foot outcomes, the latest IWGDF prevention guideline issues strong recommendations for structured education and foot self-care despite low certainty of the underlying evidence; therefore, findings should be interpreted cautiously and supported by further high-quality trials [2, 13].

Diabetic foot can be prevented or mitigated through early nursing interventions. However, a review of the literature on diabetic foot reveals that studies have predominantly focused on knowledge [7, 14, 15], attitudes [14, 16, 17], behaviors [5–7], and self-efficacy [18, 19] levels, and are mostly descriptive in nature with a limited number of interventional studies. Moreover, it is noteworthy that most interventional studies examining the effectiveness of educational interventions also included ‘diabetes-specific’ self-management education programs [20–22]. At this point, a review of the relevant literature reveals no studies that combined structured and technology-based educational methods led by nurses for diabetic foot. Therefore, educational programs aimed at developing knowledge, attitudes, and behaviors related to self-management are crucial components of effective self-management strategies. In fact, considering today’s groundbreaking information technologies, enriching these educational programs with more innovative, interactive, technology-focused, and learner-centered methods, rather than traditional methods, is no longer a choice but an inevitable necessity.

Effective management of T2DM requires not only medical treatment and behavioral changes but also self-management skills. However, traditional care models may be limited in fostering lasting behavioral change due to high patient-to-nurse ratios and time constraints. In this context, nurse-led interventions are ideal for filling this gap by holistically assessing patients, focusing on patient empowerment, and utilizing therapeutic communication skills [23, 24]. The originality of this study lies in its “hybrid” design, which integrates digital technologies (WhatsApp and YouTube) into a nurse-led self-management education program. Unlike other studies in literature, this hybrid model, based on a self-efficacy framework, aims to provide patients with continuous, flexible, and repeatable support by eliminating spatial and temporal constraints.

Bandura's self-efficacy theory, proposed in 1977, was developed to provide a unifying theory for behavioral change. This theory suggests that all individuals can be competent and successful if they have the necessary opportunities and self-efficacy to achieve their goals. It also clearly focuses on how individuals and communities can be empowered to achieve their goals [25]. Bandura states that self-efficacy is nourished by four main sources: mastery experiences, vicarious experiences, verbal persuasion, and physiological and emotional states [25, 26]. Diabetic foot self-care requires continuous behavioral engagement and the fulfilment of daily self-management tasks. Therefore, a self-efficacy-based framework provides a theoretically sound basis for designing educational interventions aimed at preventing diabetic foot development. For this reason, this model was chosen for the study to enable participants to develop behavioral change, ensure the continuity of learned behaviors, and share their experiences. To understand the impact of the intervention, it is important to distinguish between the planned training method for the study and the continuation of this training under the leadership of nurses. The "training method" (hybrid model) adopted in this study is based on the structured delivery of content using digital tools (YouTube, WhatsApp) to facilitate the learning of correct foot care behaviors, repeatable access and application, and visual modelling. The program's delivery under "nurse leadership" also emphasizes the therapeutic relationship between the nurse and the patient, individualized coaching, and ongoing feedback to promote self-efficacy and behavioral change. This study was initiated based on the assumption that the nurse's proactive and guiding role, synergistically integrated with the application of these technological tools, could lead to significant improvements in patient outcomes. Therefore, our hypothesis was that nurse-led self-efficacy-based hybrid diabetic foot self-management education program (DFS-MEP) could improve diabetic foot knowledge, foot care behaviors, adaptation to chronic illness, and self-efficacy in T2DM patients.

Methods

Study design and setting

This was a two-arm, single-center and single-blinded, randomized controlled trial. The trial was conducted between January 2023 and January 2024 at the endocrinology outpatient clinics of a tertiary-level hospital in Ankara, Türkiye.

Participants recruitment and eligibility

The study's target population comprised individuals with T2DM who had not been diagnosed with diabetic foot. Participants were recruited from routine follow-up visits at the endocrinology outpatient clinics using a sequential

sampling strategy. The inclusion criteria were as follows: (1) adults aged 18 years or older, (2) had a confirmed diagnosis of T2DM for a minimum duration of one year, (3) had an HbA1c level of 7.5% or lower [27], (4) were not receiving insulin therapy, (5) were capable of using WhatsApp and YouTube themselves or with the assistance of a relative, and (6) voluntarily provided informed consent. The subsequent criteria for exclusion were as follows: (1) pregnancy or intention to conceive, (2) a diagnosed mental or cognitive impairment, (3) a medical diagnosis of diabetic foot, and (4) a diagnosis of more severe diabetes-related comorbidities such as peripheral artery disease, retinopathy, or nephropathy.

Eligible patients were approached during their routine endocrinology outpatient clinic visits. After providing patients with verbal information about the study's purpose, procedures, duration, and expected outcomes, the researcher informed them that their participation was entirely voluntary and that they could withdraw at any time without affecting their treatment and/or care. Patients who volunteered to participate in the study were screened for eligibility and invited to participate.

Sample size

To determine the sample size and power, we utilized the G*Power 3.1.9.7 software. In a prior study [28], the effect size was determined to be 0.94 using the mean scores and standard deviations of diabetic foot self-efficacy assessments from both the intervention and control groups. Based on a two-tailed hypothesis, a significance level (α) of 0.05, and a power ($1 - \beta$) of 80%, it was determined that a minimum of 38 participants (19 per group) were required. The final sample size was adjusted to at least 50 individuals to ensure sufficient statistical power due to a projected dropout rate of 30% [29].

Randomization and blinding

Using a block randomization method with a block size of five, participants were randomly assigned to either the intervention group or the control group. To ensure that the groups were evenly distributed, ten different assignment sequences (such as AABBA, BAAAB, BABBA, etc.) were generated in advance. A separate statistician who was not involved in any other part of the study performed the randomization.

This study used a single-blind design. The person who administered the DFSMEP could not be blinded because the main investigator was responsible for this task. In the dataset, group assignments were labeled "A" and "B" to keep them blind during data analysis. To reduce bias in detection and reporting, an independent statistician who did not know which group the data belonged to conducted the analysis. Two participants in the control group were lost to follow-up due to inability to complete

the scheduled assessments, resulting in a final sample of 48 participants.

Operational definitions of variables

The independent variable of this study was the “*Nurse-Led Hybrid Diabetic Foot Self-Management Education Program*.” This concept is operationally defined as a structured 8-week intervention combining individual face-to-face training with digital follow-up (WhatsApp) and visual modeling (YouTube videos).

The study focused on four main dependent variables:

1. **Diabetic Foot Knowledge:** Defined as the participant’s level of theoretical understanding regarding foot care principles, measured by the total score obtained from the *Diabetic Foot Knowledge Scale (DFKS)*.
2. **Foot Care Behaviors:** Defined as the frequency of performing correct foot self-care practices (e.g., washing, inspecting, moisturizing), measured by the *Foot Care Behavior Scale (FCBS)*.
3. **Self-Efficacy:** Defined as the participant’s confidence in their ability to successfully organize and execute foot care tasks, measured by the *Diabetic Foot Care Self-Efficacy Scale (DFCSES)*.
4. **Adaptation to Chronic Illness:** Defined as the psychological and social adjustment of the

participant to their condition, measured by the *Adaptation to Chronic Illness Scale (ACIS)*.

Intervention

The study was conducted in two stages: development and implementation of the DFSMEP (Fig. 1). The baseline procedures for all groups were the same (Fig. 2), and follow-up tests were administered in weeks 4 and 8. The DFKS, FCBS, ACIS, and DFCSES were all administered three times: at baseline (before randomization), at week 4 (post-intervention), and at week 8 (follow-up). The diabetic foot assessment form (DFAF) was given only at baseline and again in week 8.

To prevent data contamination, participants in the intervention and control groups were scheduled to come to the clinic on different days, which kept their interactions to a minimum. The WhatsApp group and YouTube content were accessible only to participants in the intervention group. Participants in the control group did not receive any of the educational information until all follow-up data collection was completed. No participants from either group were aware of the specific procedures applied to the other arm.

Development stage

Before this study, researchers conducted a multi-stage process to develop the self-management intervention. First, a comprehensive literature review was conducted

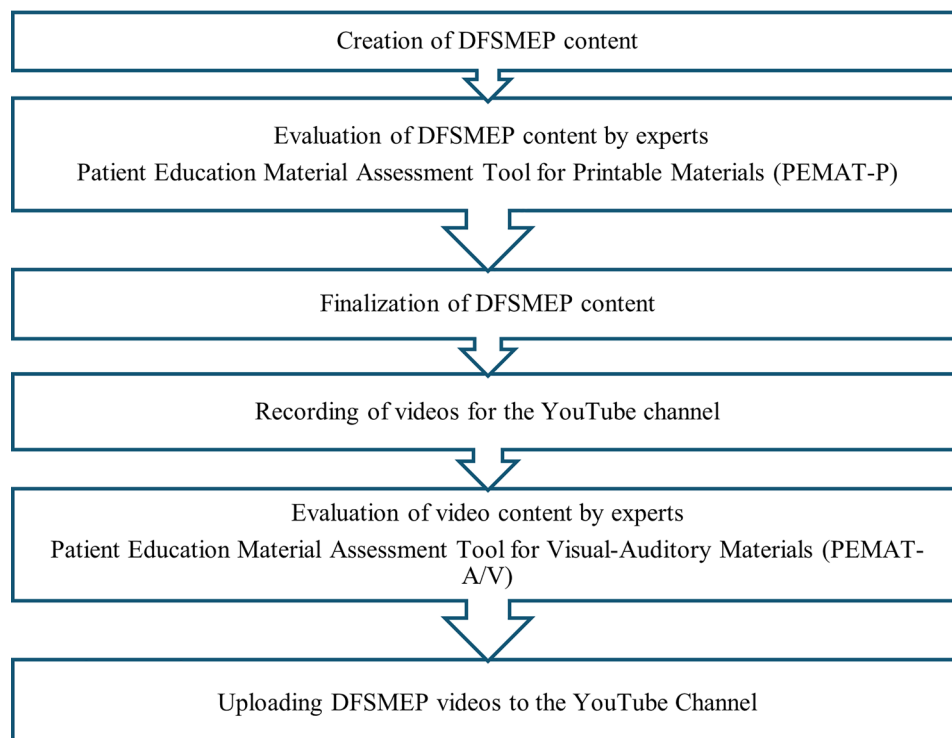


Fig. 1 Development of DFSMEP

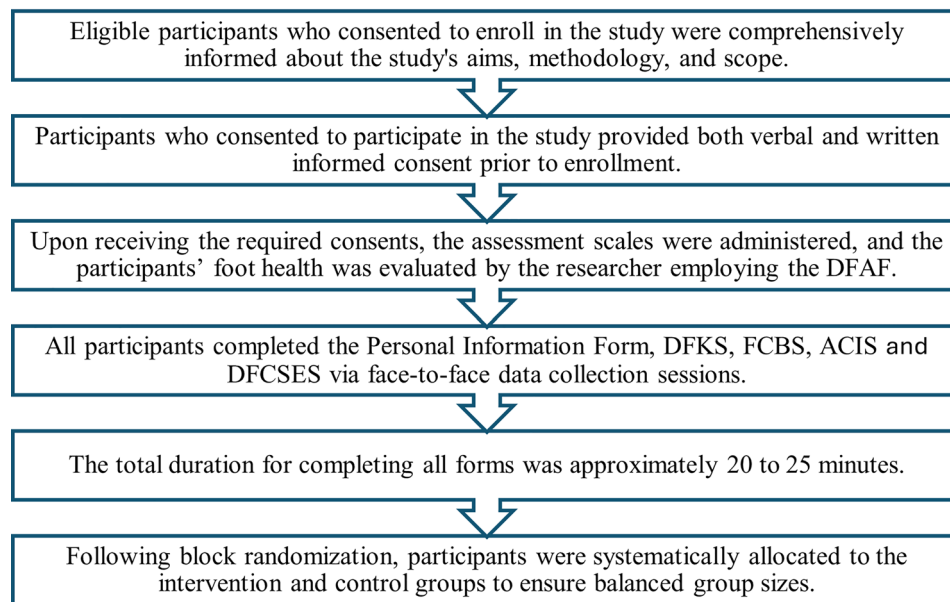


Fig. 2 Common implementation steps

to analyze foot care needs and the research status in diabetic patients. Second, the educational content and intervention were structured in line with Bandura's self-efficacy theory. To encourage "*mastery experience*", participants first observed foot examinations and care practices demonstrated by the nurse during face-to-face sessions, then performed them themselves. "*Vicarious experience*" was facilitated using YouTube videos visually demonstrating appropriate foot care and exercise practices in similar patients. "*Verbal persuasion*" was provided through the nurse's encouragement, constructive feedback, and personalized reminders sent weekly via WhatsApp. Supportive communication techniques like active listening, empathy, and validating feelings were used to address their *physiological and emotional states*, aiming to reduce concerns about foot problems and thereby increase patients' confidence in their self-care abilities. Patients were encouraged to talk about their fears, especially about diabetic foot difficulties, during face-to-face consultations and to express their concerns during follow-up calls. In line with all these components, the content included the following headings: (1) diabetes and foot health, (2) foot care, (3) nutrition, and (4) exercise. Finally, draft materials and the intervention program were reviewed by six experts, including an endocrinologist, diabetes nurses, and academicians in the field of medical nursing. Revisions were made based on expert feedback.

Evaluation of DFSMEP content

The content and information quality of the DFSMEP were evaluated using the 'Patient Education Material Assessment Tool (PEMAT)' developed by Shoemaker et

al. (2014) and validated in Turkish [30]. Since the prepared content was also provided to patients in booklet form, the expert evaluation was conducted using the Patient Education Material Evaluation Tool for printable materials (PEMAT-P). As a result of the expert evaluation, the education booklet scored 99.0% in the comprehensibility sub-dimension and 100% in the applicability sub-dimension. The content, which was revised in line with expert opinions, was recorded on video accompanied by a PowerPoint presentation and uploaded to the YouTube channel (Emine Çelik Polat) (Fig. 3).

Video recording of DFSMEP content

Following the completion of the expert evaluation of the DFSMEP, the information contained in the program was presented using PowerPoint. Video recordings were made with the researcher's audio and video presentation. With the help of physiotherapists, a video was recorded showing the exercises in practice under the heading of Physical Activity and Exercise. Specifically, the videos uploaded to the YouTube channel included a presentation explaining the effects of diabetes on foot health, foot examinations, hygiene steps, methods for cutting toenails, explanations and visuals covering footwear assessment, and exercise demonstrations guided by physiotherapists.

Evaluation of DFSMEP video recordings

The same experts evaluated the video recordings of the researcher's audio and visual presentation before being uploaded to the YouTube channel (Fig. 3). The experts conducted their evaluations using the PEMAT for audio-visual (PEMAT-A/V) materials. As a result of the expert evaluation, the videos received a score of 94.23% in the



Fig. 3 Image from powerpoint and youtube channel

comprehensibility sub-dimension and 100% in the applicability sub-dimension. The experts did not suggest any changes to the content or presentation of the videos.

Implementation of DFSMEP

DFSMEP consisted of three stages:

Stage 1: Face-to-face education at the hospital

When participants arrived at the hospital, they received education on diabetes, foot health, foot care, nutrition, and exercise. This phase included individual or group face-to-face sessions, each lasting between 15 and 40 min.

The first stage consisted of three sessions:

- **Session 1 (15 min):** Patients were given face-to-face education on healthy feet, the effects of diabetes on the feet, the concept of diabetic foot, risk factors for diabetic foot, and signs and symptoms, followed by a printed booklet containing this information. Additionally, a physical examination of the patients' feet was conducted using the Diabetic Foot Assessment Form developed by the Diabetes Nursing Association's Diabetic Foot Working Group.
- **Session 2 (30–40 min):** In this session, content related to foot care, nutrition, and exercise was shared with participants, including sample images and videos. In addition, recommendations for a healthy lifestyle to prevent diabetic foot development

and self-management skills that need to be developed were also presented to patients using a PowerPoint presentation and a handbook.

- **Session 3 (10 min):** Patients' understanding of the education program was assessed, and their questions were answered.

Stage 2: Creation of Whatsapp group and YouTube channel

After completing stage 1 with all patients in the intervention group, a WhatsApp group was created consisting only of researchers and participants. To reinforce participants' learning, video recordings were also created by the researcher's PowerPoint presentation and uploaded them to the YouTube channel (Fig. 3). To increase viewership, four separate videos were uploaded to the YouTube channel, one per week, covering the main topics in the DFSMEP content: (1) diabetes and foot health, (2) foot care, (3) nutrition, and (4) exercise. These videos were uploaded to the channel one per week, and each video link was shared with participants via the WhatsApp group at the beginning of each week (on Mondays). Participants' viewing status was tracked based on the number of views on the YouTube channel. The number of views for each video did not fall below the number of participants in the intervention group.

In addition, summaries of the information contained in the DFSMEP content were shared via the WhatsApp group twice a week (on Mondays and Fridays) for eight weeks. Participants were asked questions related to that

week's topic via the WhatsApp group after the training to keep the group active. WhatsApp group settings were configured so that only the group administrator (researcher) could send messages. However, the group was kept open for sending messages only on Mondays and Fridays between 13:00 and 16:00 to allow participants to send questions and/or feedback. No additional questions were received from participants during the times when the group was open for feedback. The researcher monitored the read status of messages sent in the group, and participants who read the messages were noted.

Stage 3: Follow-up

In the third stage, telephone calls were made to patients in the fourth and eighth weeks after the intervention, including two follow-ups. During these calls, patients were reminded of the importance of reading the necessary messages related to DFSMEP and watching and completing the videos on the YouTube channel. If a patient did not participate in these activities, they were sent a personalized (direct) message with reminders to ensure their participation. In addition, measurement tools were reapplied to the patients during the relevant weeks of this stage.

Control group

Participants in this group were given general information about pancreatic anatomy and physiology using verbal presentation techniques. This training was given only once. This intervention was independent of the content and was carried out to allow the participants to become accustomed to the clinical setting. Following completion of the study, participants were provided with a printed booklet and a YouTube channel link.

Instruments

The personal information form

The researchers created this form based on relevant literature [11, 20]. It had 20 questions about sociodemographic and clinical characteristics. There were nine demographic questions (age, gender, marital status, and education level) and 11 questions about illness and treatment-related factors (like duration of DM, type of treatment received, and DM education status).

Diabetic foot knowledge scale (DFKS)

The DFKS, a five-item subscale developed by Garcia et al. (2001), assesses foot health knowledge [31]. Items are scored as 1 (correct) or 0 (incorrect); higher scores indicate greater knowledge. The original scale yielded a Cronbach's alpha of 0.63; the Turkish version by Kır Biçer and Enç (2016) demonstrated 0.58 [28]. In this study, Cronbach's alpha was 0.47.

Foot care behavior scale (FCBS)

The FCBS, developed by Borges and Ostwald (2008) assesses foot care behavior frequency [32]. The 15-item Turkish version adapted by Kır Biçer and Enç (2016) uses a 5-point Likert scale (1 = never to 5 = always); higher scores indicate better behaviors [28]. The Turkish version reported a Cronbach's alpha of 0.83; in this study, it was 0.80.

Adaptation to chronic illness scale (ACIS)

The ACIS was originally developed in Turkish to assess adaptation to chronic illness across physical, social, and psychological domains. It comprises 25 items, with a total score range of 25–125; higher scores indicate better adaptation [33]. The original Cronbach's alpha was 0.88; in this study, it was 0.82.

Diabetic foot care self-efficacy scale (DFCSES)

The DFCSES measures confidence in diabetic foot care. It includes nine items scored on an 11-point Likert scale (0–10), with total scores ranging from 0 to 90; higher scores indicate greater self-efficacy [34]. The Turkish version reported a Cronbach's alpha of 0.86 [28]; in this study, it was 0.82.

Diabetic foot assessment form (DFAF)

The DFAF, developed by the Turkish National Diabetes Nursing Association Diabetic Foot Study Group, assesses physical foot status and care.

Data analysis

Data were analyzed using IBM SPSS version 23.0 and R statistical software. Normality of data distribution was assessed by evaluating skewness and kurtosis coefficients, with values between ± 3 considered acceptable for normal distribution. Descriptive statistics were reported as frequencies and percentages for categorical variables and as mean $\pm$ standard deviation, median (min–max), or trimmed mean $\pm$ standard error for continuous variables, depending on distribution. Independent samples t-tests were used to compare continuous variables that were normally distributed between groups. Yates' corrected chi-square, Fisher's exact, or Fisher-Freeman-Halton tests were used to compare categorical variables. Bonferroni-corrected Z-tests were employed for multiple comparisons. Repeated Measures Analysis of Variance (ANOVA) was used to compare normally distributed data across groups and over time. For non-normally distributed data, a two-way robust test was performed using the WRS2 package in R, with Bonferroni correction applied for multiple comparisons. Partial eta squared (η^2) was used to determine how clinically relevant the results were. According to accepted criteria, η^2 values of 0.01, 0.06, and 0.14 correspond to small, medium, and large

effect sizes, respectively [35, 36]. All analyses used a two-tailed significance level of $p < 0.05$.

Ethical considerations

Ethical approval for the study was obtained from the Clinical Research Ethics Committee of Health Sciences University Dışkapı Yıldırım Beyazıt Training and Research Hospital (Approval date: 29 August 2022; Registration number: 145/10). Official permission was also obtained from the institution where the study was conducted. The study was conducted in accordance with the Declaration of Helsinki and was registered at ClinicalTrials.gov (Identifier: NCT05974592; registered on July 26, 2023). Written and verbally informed consent was obtained from all participants after explaining the

study details and prior to the baseline assessment and randomization.

Results

Of the 60 patients initially assessed, four were excluded due to ineligibility, and six did not wish to participate in the study. The remaining 50 eligible volunteers were randomized into the study. Following randomization, two participants were excluded from failure to complete follow-up assessments, and the final sample consisted of 48 participants (Fig. 4).

Descriptive characteristics of the participants

Table 1 presents the descriptive characteristics of the study sample. The mean age of the participants in the intervention group was 52.68 years, with 60.0% being

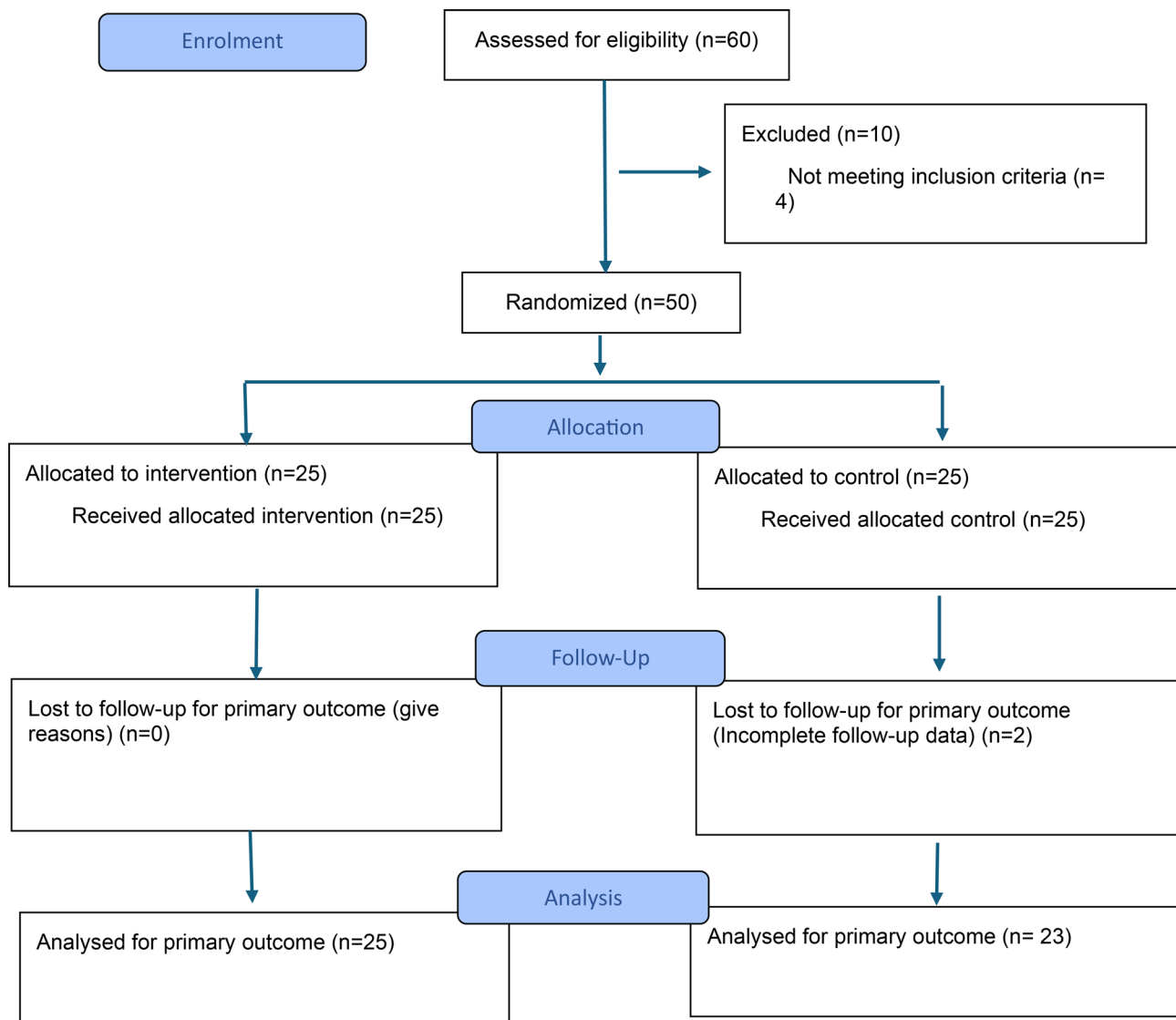


Fig. 4 CONSORT flow diagram

Table 1 Distribution of descriptive characteristics of patients in groups and comparison between groups

Variable	Group		Total	Test Statistic	p
	Intervention (n = 25)	Control (n = 23)			
Age (years) (M ± SD)	52,68 ± 12,83	55,35 ± 11,59	53,96 ± 12,2	-0.754	0.455
DM duration (years) (M ± SD)	9.66 ± 8.42	8.83 ± 10.02	9.26 ± 9.13	0.313	0.756
	n (%)	n (%)	n (%)		
Gender					
Female	15 (60,0)	18 (78,3)	33 (68,8)	1.106	0.293*
Male	10 (40,0)	5 (21,7)	15 (31,3)		
DM Education Status					
Yes	10 (40)	8 (34,8)	18 (37,5)	0.006	0.941*
No	15 (60)	15 (65,2)	30 (62,5)		
Educator of DM Education					
Nurse	6 (54,5)	9 (90)	15 (71,4)	---	0.149**
Doctor	5 (45,5)	1 (10)	6 (28,6)		
Topics Covered in Education					
Foot care education					
Yes	3 (12)	5 (21,7)	8 (16,7)	---	0.454**
No	22 (88)	18 (78,3)	40 (83,3)		
Diet/nutrition education					
Yes	12 (48)	9 (39,1)	21 (43,8)	0.107	0.743*
No	13 (52)	14 (60,9)	27 (56,2)		
Blood sugar monitoring education					
Yes	4 (16)	6 (26,1)	10 (20,8)	---	0.487**
No	21 (84)	17 (73,9)	38 (79,2)		
Medication use education					
Yes	9 (36)	8 (34,8)	17 (35,4)	0	1.000**
No	16 (64)	15 (65,2)	31 (64,6)		

X: Mean, SD: Standard Deviation, *Yates Correction; **Fisher's Exact Test

female. The mean age of the control group was 55.35 years, with 78.3% being female. In the intervention group, 60% of participants had not received any prior diabetes education. Among the 40.0% who received education, 54.5% obtained it from a nurse. Most of the education focused on diet and nutrition, accounting for 48.0%. In the control group, 65.2% of participants were had not received prior education on diabetes. Among those who had received education, 34.2% obtained it from a nurse, who was the main provider. Diet and nutrition were the major topics of discussion, accounting for 39.1%.

Effects on diabetic foot knowledge

A statistically significant difference was noted between the intervention and control groups regarding DFKS total scores ($p < 0.001$), with the intervention group attaining higher scores at all time points (Table 2). A notable improvement over time was recorded between the baseline and later evaluations ($p < 0.001$). The trimmed-mean DFKS scores were 3.55 at baseline, 4.27 in week 4, and 4.39 in week 8 (Fig. 5). A notable Group $\times$ Time interaction effect was also found ($p < 0.001$). The highest trimmed mean score in the intervention group was recorded in week 8 (5.00), whereas the minimum score

was noted at baseline in the intervention group (3.48) and in week four in the control group.

Effects on foot care behavior

The results indicated significant differences in FCBS scores in terms of group ($F = 8.57$, $p = 0.004$, $\eta^2 = 0.058$), time ($F = 13.32$, $p < 0.001$, $\eta^2 = 0.162$), and Group $\times$ Time interaction ($F = 12.80$, $p < 0.001$, $\eta^2 = 0.157$). When analyzed by group, the intervention group had a mean total FCBS score of 64.16, while the control group had a mean score of 59.87. Over time, the mean total FCBS score for the entire sample improved from 56.54 at baseline to 64.96 at week 4 and 64.81 at week 8 (Fig. 6). The Group $\times$ Time interaction effect was large ($\eta^2 = 0.157$), demonstrating a considerable effect of the DFSMEP on foot care behaviors over time. The intervention group exhibited the greatest mean score in week 4 (70.20), whereas the lowest was noted at baseline in the same group (53.64).

Effects on adaptation to chronic illness

No statistically significant differences were found in median ACIS total scores regarding group ($p = 0.194$), time ($p = 0.988$), or Group $\times$ Time interaction ($p = 0.965$) (Fig. 7).

Table 2 Knowledge, foot care, adherence and self-efficacy scores before and after intervention by group

Variables	Time	Intervention (n=25) M ± SD	Control (n=23) M ± SD	Total	Source	F	p ^a	η ²
DFKS *	Baseline	3,48 ± 0,16 ^A	3,67 ± 0,18 ^A	3,55 ± 0,11 ^b 4,27 ± 0,15 ^a 4,39 ± 0,12 ^a	Group	66,100	0,001	-
	4. week	4,91 ± 0,07 ^B	3,48 ± 0,17 ^A		Time	34,100	0,001	
	8. week	5,00 ± 0,00 ^C	3,67 ± 0,11 ^A		Group*Time	37,500	0,001	
	Total	4,51 ± 0,1	3,6 ± 0,1	4,08 ± 0,08				
FCBS **	Baseline	53,64 ± 10,34 ^B	59,7 ± 10,21 ^B	56,54 ± 10,62 ^b	Group	8,57	0,004	0,058
	4. week	70,2 ± 4,62 ^A	59,26 ± 10,32 ^B	64,96 ± 9,55 ^a 64,81 ± 8,87 ^a	Time	13,32	< 0,001	0,162
	8. week	68,64 ± 5,45 ^A	60,65 ± 10,05 ^B		Group*Time	12,8	< 0,001	0,157
	Total	64,16 ± 10,38	59,87 ± 10,06	62,1 ± 10,42				
DFCSES ***	Baseline	73,36 ± 14,42 ^B	72,91 ± 17,13 ^B	73,15 ± 15,61 ^b	Group	16,98	< 0,001	0,110
	4. week	88,4 ± 10,73 ^A	73,48 ± 17,27 ^B	81,25 ± 15,97 ^a 81,81 ± 15,09 ^a	Time	5,00	0,008	0,068
	8. week	88,76 ± 9,21 ^A	74,26 ± 16,71 ^B		Group*Time	3,87	0,023	0,053
	Total	83,51 ± 13,58	73,55 ± 16,8	78,74 ± 15,95				
ACIS ****	Baseline	98 (68–125)	94 (64–119)	95 (64–125) 95 (64–121) 95 (64–121)	Group	1,686	0,194	-
	4. week	98 (68–121)	95 (64–119)		Time	0,013	0,988	
	8. week	98 (68–121)	94 (64–119)		Group*Time	0,071	0,965	
	Total	98 (68–125)	94 (64–119)	95 (64–125)				

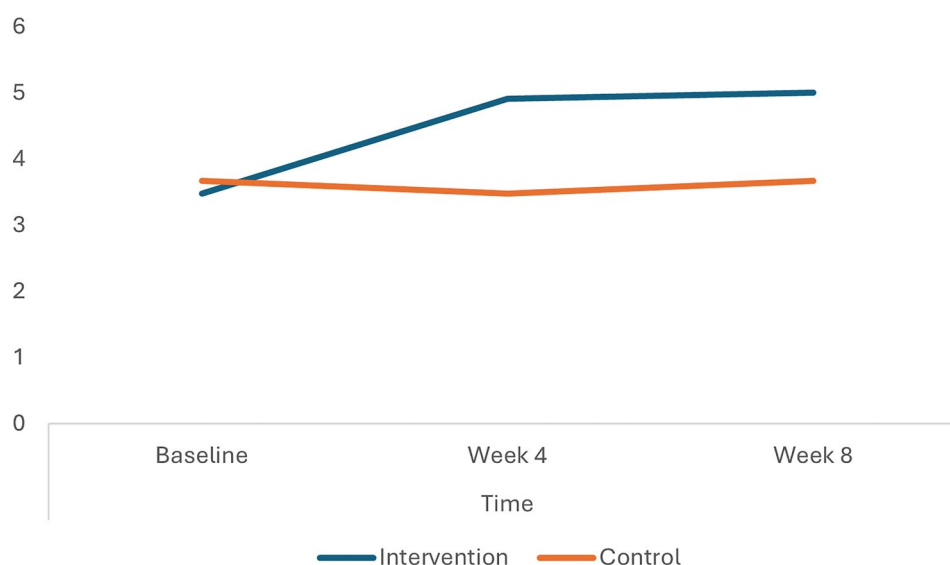
*Robust ANOVA; Trimmed mean method used in comparisons. Tests based on a 20% trimmed mean were applied in comparisons. Therefore, the partial eta squared (η_p^2) value, which is the classic effect size measure, was not calculated

a-b: No significant difference between groups sharing the same letter; A-C: No significant interaction differences sharing the same letter

**F: Generalized Linear Models; $R^2=31.36\%$; Adjusted $R^2=28.88\%$; η^2 : Partial Eta Squared

***F: Generalized Linear Models; $R^2=20.46\%$; Adjusted $R^2=17.58\%$

****Robust ANOVA; Median method used in comparisons. Median (Minimum–Maximum)

**Fig. 5** Comparison of diabetic foot knowledge scores (DFKS)

Despite the lack of statistical significance, adaptation levels in the intervention group exhibited stability over time, indicating possible therapeutic relevance that necessitates additional exploration.

Effects on diabetic foot care self-efficacy

The DFCSES total scores demonstrated significant differences in terms of group ($F=16.98$, $p<0.001$, $\eta^2 = 0.110$), time ($F=5.00$, $p=0.008$, $\eta^2 = 0.162$), and Group $\times$ Time interaction ($F=3.87$, $p=0.023$, $\eta^2 = 0.053$). The

intervention group showed a higher mean DFCSES total score (83.51) compared to the control group (73.55). The mean DFCSES score rose from 73.15 at baseline to 81.25 at week 4 and 81.81 at week 8 (Fig. 8). The Group $\times$ Time interaction effect was small to medium ($\eta^2 = 0.053$), indicating that the DFSMEP enhanced self-efficacy in diabetic foot care over time. The intervention group exhibited the highest mean score at week 8 (88.76), whereas the control group recorded the lowest score at baseline (72.91).

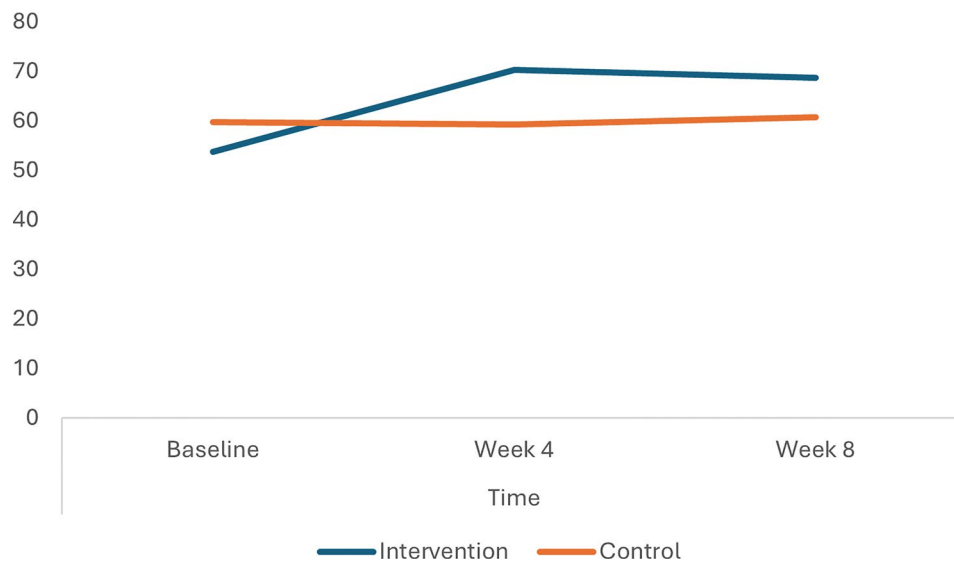


Fig. 6 Comparison of foot care behavior scores (FCBS)

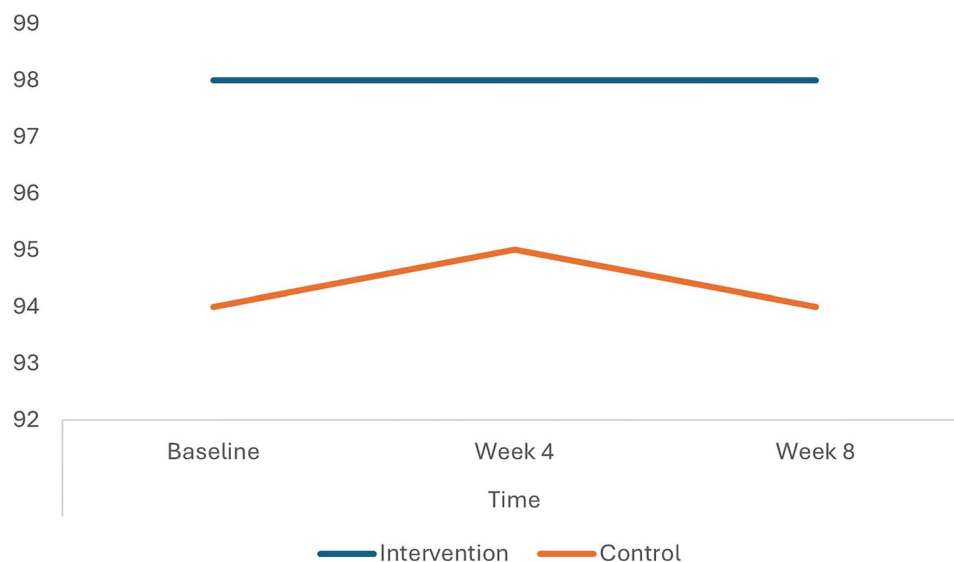


Fig. 7 Comparison of adaptation to chronic illness scale (ACIS)

Discussion

Our study is among the first to examine the effects of a self-efficacy-based nurse-led hybrid foot self-management education program on diabetic foot knowledge, foot care behaviors, adaptation to chronic illness, and self-efficacy of patients with T2DM. The study showed that the DFSMEP increased diabetic foot care knowledge, enhanced foot care behaviors and self-efficacy levels, and contributed to keeping these scores over time. However, it did not create a significant difference in adaptation to chronic illness levels between groups.

A serious consequence of DM, diabetic foot raises mortality and morbidity rates, causes extremity amputations, requires continuing medical care, and places heavy financial burdens on people and healthcare systems [37]. In a

cross-sectional study of 375 people with type 2 DM in Iran, 84.8% of patients did not have a good understanding of how to care for their feet. There was a positive link between knowledge and practice scores [38]. Similarly, Gökdeniz and Akgün Şahin (2022) found that patients with DM had a moderate knowledge about diabetic foot. Consistent with previous studies, the current study found that both groups had a moderate level of knowledge of diabetic foot care based on baseline assessments [39].

No previous study has precisely integrated face-to-face education with hybrid assistance approaches as executed in this study, nevertheless, other studies have assessed the efficacy of face-to-face and/or mobile interventions independently [40–43]. Rahaman et al. (2018) found significant improvements in foot care knowledge among

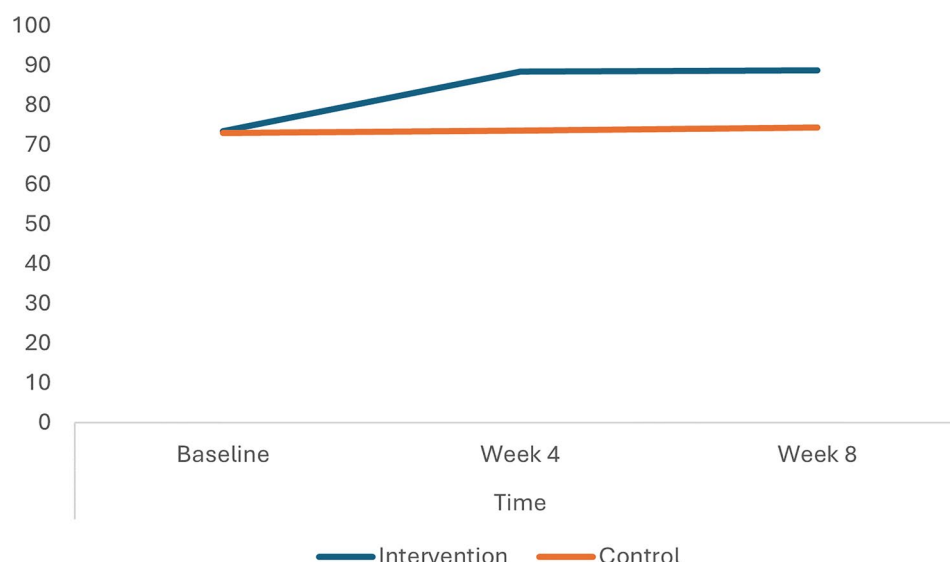


Fig. 8 Comparison of diabetic foot care self-efficacy scale scores (DFCSES)

patients receiving outpatient education at the first and third months of follow-up [40]. Similarly, Can Çiçek and Gökdoğan (2021) also noted a sustained enhancement in diabetic foot knowledge among patients with DM following a planned educational and follow-up intervention [41]. Additional research, such as that conducted by Toygar et al. (2022) and a systematic review by Yıldırım Ayaz et al. (2022), corroborates the conclusion that structured diabetic foot education improves knowledge, self-efficacy, and behavior [42, 43]. In the current study, while DFKS scores were similar between the intervention and control groups at baseline, there was a significant difference between the groups in the fourth- and eighth-week measurements. According to these results, it was determined that the nurse-led DFSMEP implemented with the intervention group effectively enhanced their understanding of diabetic foot care. The intervention group's knowledge scores increased over the subsequent weeks, attributed to the education being supplemented not just by in-person sessions but also by reinforcement via WhatsApp messages and YouTube videos. This strategy facilitated frequent exposure, accessibility, reinforcement, and memory retention.

In addition to knowledge, promoting appropriate foot care behaviors is a critical goal of diabetic foot education interventions. Education plays a key role not only in increasing knowledge but also in translating it into consistent self-care practices [4]. Structured education programs are emphasized for encouraging individuals with diabetes to adopt correct foot care practices [2]. In a quasi-experimental study, the intervention group participated in weekly 15-minute diabetes self-management sessions for eight weeks and found that the education significantly improved foot care practices, changing

them 12.2% [44]. Dincer and Bahçecik (2021) assessed the efficacy of a mobile diabetic foot care teaching program enhanced with animation [45]. The diabetic foot care behavior scores significantly rose from 52 at baseline to 63 one-month post-intervention. Conversely, the control group's score markedly declined from 52 to 43 within the same interval. A meta-analysis found that education improved foot care practices and glycemic control [46]. While there is strong evidence that health education improves foot care behaviors in patients with DM, evidence supporting its direct effect on preventing DFU recurrence remains limited, as noted in current guidelines. Thus, further studies on the long-term efficacy of diabetic foot care education programs are needed to determine the best method [13]. Similar to existing studies, this study revealed that foot care behaviors increased significantly in the intervention group, while no change was observed in the control group. An in-depth analysis of the data indicated that the substantial effect of the DFSMEP on foot care behaviors commenced in the fourth week and persisted until the eighth week. These data indicate that the efficacy of the DFSMEP in enhancing and maintaining foot care behavior scores is due to the utilization of written, visual, and auditory instructional resources, complemented by messages and videos that are available at any time and location. Furthermore, allowing participants to communicate with the researcher and ask questions undoubtedly enhanced the program's success. These findings should also be interpreted within the cultural and demographic context of the Turkish adult population with T2DM. Patient education programs for diabetes in Turkey often focus on nutrition and medication management rather than specific foot care training. This situation may explain why significant gains

in knowledge and behavior can be achieved when structured and accessible education is provided.

In addition to knowledge and behaviors, adaptation to chronic illness is a key factor influencing long-term self-management. Although chronic disease adaptation is crucial in preventing diabetic foot complications [47], no significant changes were found in ACIS scores between groups. Other studies have reported similar findings regarding moderate compliance among individuals with DM [48, 49]. The absence of significant change in adaptation in this study might be explained by initially good compliance levels among participants and the relatively short intervention duration, considering that behavioral adaptation often requires longer periods. Adapting to a chronic illness is a more difficult and complex psychological process that is affected by the patient's long-term experience with the illness, cultural acceptance, and coping strategies. The fact that our study group has been living with diabetes for a long time and reported moderate to good adaptation at baseline indicates that these patients' perceptions of the disease have already reached a certain stability. In this context, the 8-week intervention period may have been too short to change psychological adaptation patterns to a statistically significant degree, even though it was sufficient for the development of concrete behaviors. Furthermore, chronic illnesses are sometimes met with a fatalistic beliefs in Turkish culture. This may delay the psychological acceptance necessary for adaptation to chronic illness, even if patients demonstrate the necessary physical care behaviors in accordance with the nurse's instructions. Nevertheless, the fact that adjustment scores did not decline but were maintained can be considered a clinically important finding, indicating that the intervention supported patients' psychological resilience.

Self-efficacy is a critical determinant of self-care performance and significantly impacts foot care behaviors. A study examining foot care education interventions based on Bandura's theory found that they were effective in improving self-efficacy, diabetic foot knowledge, health status, and quality of life [42]. A multicenter randomized controlled trial examining the effectiveness of a self-efficacy-focused structured education program for adults with T2DM, which included general information about diabetes, diet, physical activity, and regular check-ups, found that self-efficacy and self-management behaviors improved significantly in the intervention group [50]. Although our study was not conducted using the exact same method and content as those in the literature, it was consistent with related studies in that the mean self-efficacy scores were similar in both groups at baseline but showed a significant increase in the intervention group at week eight, while no change was observed in the control group. The tendency to continue self-care behaviors with

DFSMEP may be related to the program's ability to reinforce the correct information learned through education, as well as its inclusion of written, visual, and auditory educational materials supported by messages and videos designed to enhance individuals' self-care behaviors and self-efficacy levels. Unlike other studies and the current study, in a study evaluating the effectiveness of a mobile diabetic foot personal care system, the intervention group showed a significant increase in their level of knowledge about foot care compared to the control group. However, there was no significant difference between the groups in terms of foot care behaviors and self-efficacy scores. A recent meta-analysis examining the effects of foot care education for diabetic patients on knowledge, self-efficacy, and behavior showed that knowledge and behavior levels increased, but self-efficacy was not affected [43]. It is thought that these results may have arisen due to differences in the sample groups in the studies, as well as variations in the content, duration, and method of implementation of the educational interventions.

The meaningful results emerging from the present study regarding diabetic foot knowledge, foot care behaviors, and self-efficacy can also be considered within the cultural context of the study population. For example, in settings where healthcare services are provided in Turkey, nurses often play a leading role as accessible and reliable sources of health information, particularly in chronic disease management. The intervention design of the current study, which was nurse-led and included repeated reinforcement through digital platforms, may have helped participants become more interested in and confident in foot care routines. Also, the fact that people use mobile messaging apps like WhatsApp a lot in their daily lives may have made the educational content easier to understand and follow.

There are several limitations to be considered when interpreting the findings of this study. First, the study was conducted only with Type 2 DM patients who were treated and monitored at a single center. Therefore, the results are sample-specific and cannot be generalized to all Type 2 DM patients. Second, this study examined patient outcomes after intervention only in the fourth and eighth weeks through repeated measurements. Longer-term follow-up studies are needed to investigate whether DFSMEP has lasting effects on patient outcomes.

Conclusion

This randomized controlled study demonstrated that the nurse-led hybrid DFSMEP improved diabetic foot care knowledge, foot care behaviors, and self-efficacy levels in patients with T2DM. The program did not create a significant effect on adaptation to chronic illness, but it helped maintain adaptation levels over time in the intervention

group. These results underscore the importance of nurses leading structured, technology-enhanced educational sessions to assist individuals with diabetes in managing their health. Such approaches could be integrated into standard clinical practice and national diabetic care programs to assist patients and prevent foot complications. Further research is recommended with long term follow-up periods and larger sample sizes to explore the long-term effects of such interventions on chronic disease adaptation, quality of life, and healthcare utilization. Future multicenter randomized controlled trials could enhance the database for clinical guidelines and policy decisions about diabetes management.

Implications for stakeholders

Implications for nurses and diabetes educators

DFSMEP can be used as a model that supports behavioral change and self-efficacy in diabetic foot care and can be easily integrated into routine clinical practice. This hybrid structure, enriched with digital tools (e.g. WhatsApp), makes patient follow-up more efficient in terms of time and cost, while the interactive process, supported by visual and auditory materials, can increase knowledge retention and patient motivation.

Implications for healthcare institutions and administrators

This model does not require any staff or infrastructure; therefore, it may be used as a cost-effective strategy to reduce the long-term burden of diabetic foot problems.

Implications for patients and families

This model makes it possible for patients in similar groups to have the same dependable information and support outside of the hospital, no matter where or when they are.

Implications for policy makers and health systems

National diabetes guidelines can suggest nurse-led, technology-supported teaching models as a standard of care. Implementing similar models could be an effective approach for preventive health initiatives designed to diminish morbidity and healthcare costs.

Implications for future research

Future studies should investigate the long-term efficacy of nurse-led hybrid education programs using extended follow-up durations and multi-center methodologies.

Abbreviations

ACIS	Adaptation to Chronic Illness Scale
DFKS	Diabetic Foot Knowledge Scale
DFCSES	Diabetic Foot Care Self-Efficacy Scale
DFSMEP	Diabetic Foot Self-Management Education Program
DFAF	Diabetic Foot Assessment Form
DFU	Diabetic Foot Ulcer
DM	Diabetes Mellitus

FCBS	Foot Care Behavior Scale
HbA1c	Glycated Hemoglobin
PEMAT	Patient Education Materials Assessment Tool
PEMAT-P	PEMAT for printable materials
PEMAT-A/V	PEMAT for audiovisual materials
RCT	Randomised Controlled Trial
SPSS	Statistical Package for the Social Sciences
T2DM	Type 2 Diabetes Mellitus
η^2	Partial eta squared

Acknowledgements

The authors would like to thank all patients who participated in this study.

Author contributions

Emine Çelik Polat: Conceptualization, Methodology, Data curation, Formal analysis, Investigation, Writing – original draft, Writing – review & editing. Ayşe Arıkan Dönmez: Conceptualization, Methodology, Supervision, Writing – review & editing, Project administration. Erman Çakal: Methodology, Supervision, Writing – review & editing, Project administration.

Funding

This study did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to ethical restrictions and protection of participant confidentiality, but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Ethical approval for the study was obtained from the Clinical Research Ethics Committee of Health Sciences University Dışkapı Yıldırım Beyazıt Training and Research Hospital (Approval date: 29 August 2022; Registration number: 145/10). Official permission was also obtained from the institution where the study was conducted. The study was conducted in accordance with the Declaration of Helsinki and was registered at ClinicalTrials.gov (Identifier: NCT05974592; registered on July 26, 2023). Written and verbally informed consent was obtained from all participants before and after explaining the study details and prior to the baseline assessment and randomization.

Statistical analysis statement

a. We have checked to make sure that our submission conforms as applicable to the Journal's statistical guidelines described here. b. The statistics were checked by the e-statistics platform (<https://app.eistatistik.com/>) before submission. c. We affirm that the methods used in the data analyses are suitably applied to our data within our study design and context, and the statistical findings have been implemented and interpreted correctly. d. We agree to take responsibility for ensuring that the choice of statistical approach is appropriate and is conducted and interpreted correctly as a condition to submit to the Journal.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 1 October 2025 / Accepted: 2 February 2026

Published online: 10 February 2026

References

1. Bajaj M, McCoy RG, Balapattabi K, Bannuru RR, Bellini NJ, Bennett AK, et al. 12. Retinopathy, Neuropathy, and foot care: standards of care in Diabetes—2026. *Diabetes Care*. 2026;49:S261–76. <https://doi.org/10.2337/dc26-S012>.
2. Schaper NC, van Netten JJ, Apelqvist J, Bus SA, Fitridge R, Game F, et al. Practical guidelines on the prevention and management of diabetes-related foot

- disease (IWGDF 2023 update). *Diabetes Metab Res Rev.* 2024;40. <https://doi.org/10.1002/dmrr.3657>.
3. Costa IG, Camargo-Plazas P. The impact of diabetic foot ulcer on individuals' lives and daily routine. *J Wound Ostomy Cont Nurs.* 2023;50:73–7. <https://doi.org/10.1097/WON.0000000000000941>.
 4. Alshammari L, O'Halloran P, McSorley O, Doherty J, Noble H. The effectiveness of foot care educational interventions for people living with diabetes mellitus: an umbrella review. *J Tissue Viability.* 2023;32:406–16. <https://doi.org/10.1016/j.jtv.2023.06.001>.
 5. Sari Y, Upoyo AS, Isworo A, Taufik A, Sumeru A, Anandari D, et al. Foot self-care behavior and its predictors in diabetic patients in Indonesia. *BMC Res Notes.* 2020;13:38. <https://doi.org/10.1186/s13104-020-4903-y>.
 6. Shi W, Zhou Q, Yang D, Yang J, Liu C, Pan H. Status and influencing factors of foot care behavior for patients with diabetic foot amputation: A cross-Sectional study. *Int J Low Extrem Wounds.* 2025;24:109–16. <https://doi.org/10.1177/15347346241245165>.
 7. Yücel F, Sunay D. Assessment of knowledge, attitudes and behaviours of diabetic patients about diabetic foot and foot care. *Ankara Med J.* 2016;16. <https://doi.org/10.17098/amj.53438>.
 8. Powers MA, Bardsley J, Cypress M, Duker P, Funnell MM, Fischl AH, et al. Diabetes Self-management education and support in type 2 diabetes. *Diabetes Educ.* 2017;43:40–53. <https://doi.org/10.1177/0145721716689694>.
 9. Powers MA, Bardsley JK, Cypress M, Funnell MM, Harms D, Hess-Fischl A, et al. Diabetes self-management education and support in adults with type 2 diabetes: a consensus report of the American diabetes association, the association of diabetes care & education Specialists, the academy of nutrition and Dietetics, the American academy of family Physicians, the American academy of PAs, the American association of nurse Practitioners, and the American pharmacists association. *Diabetes Care.* 2020;43:1636–49. <https://doi.org/10.2337/dci20-0023>.
 10. Sajith R, Ackers L, Ackers-Johnson S, Parker DJ, Stephens M. The practice, nature, and impact of nurse-led type 2 diabetic foot prevention services and educational programs in Sub-Saharan africa: a scoping review. *Front Public Health.* 2024;12. <https://doi.org/10.3389/fpubh.2024.1465750>.
 11. Ju H-H, Momin R, Cron S, Jularbal J, Alford J, Johnson C. A nurse-led tele-health program for diabetes foot care: feasibility and usability study. *JMIR Nurs.* 2023;6:e40000. <https://doi.org/10.2196/40000>.
 12. Subrata SA, Phuphaibul R, Grey M, Siripitayakunkit A, Piaseu N. Improving clinical outcomes of diabetic foot ulcers by the 3-month self- and family management support programs in indonesia: A randomized controlled trial study. *Diabetes Metabolic Syndrome: Clin Res Reviews.* 2020;14:857–63. <https://doi.org/10.1016/j.dsx.2020.05.028>.
 13. Goodall RJ, Ellauzi J, Tan MKH, Onida S, Davies AH, Shalhoub J. A systematic review of the impact of foot care education on self efficacy and self care in patients with diabetes. *Eur J Vasc Endovasc Surg.* 2020;60:282–92. <https://doi.org/10.1016/j.ejvs.2020.03.053>.
 14. Jia H, Wang X, Cheng J. Knowledge, attitudes, and practices associated with diabetic foot prevention among rural adults with diabetes in North China. *Front Public Health.* 2022;10. <https://doi.org/10.3389/fpubh.2022.876105>.
 15. Ramirez-Perdomo C, Perdomo-Romero A, Rodríguez-Vélez M. Knowledge and practices for the prevention of the diabetic foot. *Rev Gaucha Enferm.* 2019;40:e20180161. <https://doi.org/10.1590/1983-1447.2019.20180161>.
 16. Alharbi MO, Sulaiman AA. Foot care knowledge, attitude and practices of diabetic patients. *J Family Med Prim Care.* 2022;11:3816–23. https://doi.org/10.4103/jfmpc.jfmpc_183_21.
 17. Alshaikh AA, Alqarni HM, Hassan Assiri HA, Shlwan MA, AlJebreel MA, Almuaddi AS, et al. Knowledge, attitude, and practice of diabetic foot ulcer care in Asser region: a cross-sectional study. *Cureus.* 2023. <https://doi.org/10.7759/cureus.42807>.
 18. Sezgunsay E, Urkan M, Deveci M. Diabetic foot care behavior and self-efficacy levels in individuals with diabetic foot ulcers in Turkey. *J Tissue Viability.* 2025;34:100885. <https://doi.org/10.1016/j.jtv.2025.100885>.
 19. Dağcı S, Ören B, Özşenel EB. An examination of the self-efficacy and factors influencing foot care behaviors in individuals with type 2 diabetes mellitus. *Int J Low Extrem Wounds.* 2024. <https://doi.org/10.1177/15347346241296961>.
 20. Cengiz D, Korkmaz F. Effectiveness of a nurse-led personalized patient engagement program to promote type 2 diabetes self-management: A randomized controlled trial. *Nurs Health Sci.* 2023;25:571–84. <https://doi.org/10.1111/nhs.13048>.
 21. Ferreira G, Carvalho A, Pereira MG. Educational interventions on diabetic foot Self-Care: A study protocol for a pragmatic randomized controlled trial. *Health Commun.* 2024;39:1383–92. <https://doi.org/10.1080/10410236.2023.213875>.
 22. Ergolu N, Sabuncu N. The effect of education given to type 2 diabetic individuals on diabetes self-management and self-efficacy: randomized controlled trial. *Prim Care Diabetes.* 2021;15:451–8. <https://doi.org/10.1016/j.pcd.2021.02.011>.
 23. Sun J, Fan Z, Kou M, Wang X, Yue Z, Zhang M. Impact of nurse-led self-management education on type 2 diabetes: a meta-analysis. *Front Public Health.* 2025;13. <https://doi.org/10.3389/fpubh.2025.1622988>.
 24. Tamiru S, Dugassa M, Amsalu B, Bidira K, Bacha L, Tsegaye D. Effects of Nurse-Led diabetes Self-Management education on Self-Care knowledge and Self-Care behavior among adult patients with type 2 diabetes mellitus attending diabetes follow up clinic: A Quasi-Experimental study design. *Int J Afr Nurs Sci.* 2023;18:100548. <https://doi.org/10.1016/j.ijans.2023.100548>.
 25. Bandura A. Self-efficacy: toward a unifying theory of behavioral change. *Psychol Rev.* 1977;84:191–215. <https://doi.org/10.1037/0033-295X.84.2.191>.
 26. Gallagher MW. Self-Efficacy. *Encyclopedia of Human Behavior*, Elsevier; 2012, pp. 314–20. <https://doi.org/10.1016/B978-0-12-375000-6.00312-8>.
 27. National Institute for Health and Care Excellence. Type 2 diabetes in adults: Management (NICE guideline NG28). 2022.
 28. Kır Bıçer E, Enç N. Evaluation of foot care and self-efficacy in patients with diabetes in turkey: an interventional study. *Int J Diabetes Dev Ctries.* 2016;36:334–44. <https://doi.org/10.1007/s13410-016-0464-y>.
 29. Iqhrammullah M, Yudhistira Refin R, Fitria Andika F, Amirah S, Fahd Abdurrahman M, Alina M, et al. Dropout rate in clinical trials of smartphone apps for diabetes management: A meta-analysis. *Diabetes Res Clin Pract.* 2024;212:111723. <https://doi.org/10.1016/j.diabres.2024.111723>.
 30. Paylan Akkoç C, Orgun F. Psychometric testing of the Turkish version of the patient education materials assessment tool. *Florence Nightingale J Nurs.* 2023;31:180–7. <https://doi.org/10.5152/FNJN.2023.22196>.
 31. Garcia AA, Villagomez ET, Brown SA, Kouzekanani K, Hanis CL. The Starr County diabetes education study. *Diabetes Care.* 2001;24:16–21. <https://doi.org/10.2337/diacare.24.1.16>.
 32. Borges WJ, Ostwald SK. Improving foot Self-Care behaviors with pies Sanos. *West J Nurs Res.* 2008;30:325–41. <https://doi.org/10.1177/0193945907303104>.
 33. Atik D, Karatepe H. Scale development study: adaptation to chronic illness. *Acta Med Mediterranea.* 2016;32:135–42.
 34. Quarles BE. Educational methods increasing self-efficacy for the management of foot care in adults with diabetes and implementation of foot care behaviors. 2005.
 35. Lakens D. Calculating and reporting effect sizes to facilitate cumulative science: a practical primer for t-tests and ANOVAs. *Front Psychol.* 2013;4. <https://doi.org/10.3389/fpsyg.2013.00863>.
 36. Morris PE, Fritz CO. Effect sizes in memory research. *Memory.* 2013;21:832–42. <https://doi.org/10.1080/09658211.2013.763984>.
 37. Armstrong DG, Tan T-W, Boulton AJM, Bus SA. Diabet Foot Ulcers. *JAMA.* 2023;330:62. <https://doi.org/10.1001/jama.2023.10578>.
 38. Pourkazemi A, Ghanbari A, Khojamli M, Balo H, Hemmati H, Jafaryparvar Z, et al. Diabetic foot care: knowledge and practice. *BMC Endocr Disord.* 2020;20:40. <https://doi.org/10.1186/s12902-020-0512-y>.
 39. Gökdeniz D, Akgün Şahin Z. Evaluation of knowledge levels about diabetes foot care and self-care activities in diabetic individuals. *Int J Low Extrem Wounds.* 2022;21:65–74. <https://doi.org/10.1177/1534734620926266>.
 40. Rahaman H, Jyotsna V, Sreenivas V, Krishnan A, Tandon N. Effectiveness of a patient education module on diabetic foot care in outpatient setting: an open-label randomized controlled study. *Indian J Endocrinol Metab.* 2018;22:74. https://doi.org/10.4103/ijem.IJEM_148_17.
 41. CAN ÇİÇEKS. Effect of education and monitoring on developing foot care of elderly with diabetes mellitus. *Clin Experimental Health Sci.* 2021;11:471–80. <https://doi.org/10.33808/clinexphthsci.815251>.
 42. Toygar İ, Hançerlioğlu S, Utku T, Şimşir İY, Çetinkalp Ş. Effect of an educational intervention based on bandura's theory on foot care Self-Efficacy in diabetes: A prospective Quasi-Experimental study. *Int J Low Extrem Wounds.* 2022;21:414–9. <https://doi.org/10.1177/1534734620948327>.
 43. Yıldırım Ayaz E, Dincer B, Oğuz A. The effect of foot care education for patients with diabetes on Knowledge, Self-Efficacy and behavior: systematic review and Meta-Analysis. *Int J Low Extrem Wounds.* 2022;21:234–53. <https://doi.org/10.1177/15347346221109047>.
 44. Frisca S. Effectiveness diabetes Self-Management education (DSME) to foot care behaviour and foot condition in diabetes mellitus patient. *KnE Life Sci.* 2021;10:69–77. <https://doi.org/10.18502/kl.v6i1.8783>

45. Dincer B, Bahçecik N. The effect of a mobile application on the foot care of individuals with type 2 diabetes: A randomised controlled study. *Health Educ J*. 2021;80:425–37. <https://doi.org/10.1177/0017896920981617>.
46. Lira JAC, Rocha ASC, Bezerra SMG, Nogueira PC, dos Santos AMR, Nogueira LT. Effects of educational technologies on the prevention and treatment of diabetic ulcers: A systematic review and meta-analysis. *Rev Lat Am Enfermagem* 2023;31. <https://doi.org/10.1590/1518-8345.6628.3944>
47. Kardaş Kin Ö, Türeyen A. The effect of diabetes education based on learning modality in individuals with diabetes incompatible with treatment on compliance and metabolic goals: A randomized controlled trial. *Prim Care Diabetes*. 2022;16:150–5. <https://doi.org/10.1016/j.pcd.2021.12.006>.
48. Marinho FS, Moram CBM, Rodrigues PC, Leite NC, Salles GF, Cardoso CRL. Treatment adherence and its associated factors in patients with type 2 diabetes: results from the Rio de Janeiro type 2 diabetes cohort study. *J Diabetes Res*. 2018;2018:1–8. <https://doi.org/10.1155/2018/8970196>.
49. Çorak B, Uysal N. The effect of health perception and family support on the level of adherence to treatment in individuals with type 2 diabetes mellitus. *Med J Ankara Train Res Hosp*. 2022;55:125–9. <https://doi.org/10.20492/aeahtd.1098831>.
50. Jiang X, Jiang H, Lu Y, Liu S, Wang J, Tang R, et al. The effectiveness of a self-efficacy-focused structured education programme on adults with type 2 diabetes: A multicentre randomised controlled trial. *J Clin Nurs*. 2019;28:3299–309. <https://doi.org/10.1111/jocn.14908>.

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